Course: Internet of Things (IoT)
Assignment: IoT Final Project Proposal
Submission: Before starting practical implementation

Instructions

Each group is required to prepare an IoT project proposal based on a real-life problem they would like to solve. The proposed project should involve an microcontroller, sensors or other electronic devices, and data communication. The proposal must include the following sections:

1. **Project Title:** Give your project a clear and descriptive title.
2. **Problem Statement:** Explain the problem your project intends to solve and why it is important.
3. **Project Objectives:** State the main objective and specific objectives of the project.
4. **Proposed Solution:** Describe how your IoT system will work, from data collection to communication, processing, and visualization.
5. **Required Devices and Materials:** Provide a detailed list of all the devices, electronic components, software, and other materials needed to implement the project. Include the quantity and purpose of each item.
6. **System Architecture:** Draw a block diagram showing how the sensors, microcontroller, communication protocol, server, and other components will communicate.
7. **Expected Results:** Explain what the system is expected to achieve.
8. **Estimated Budget:** Provide the estimated cost of the required hardware and materials in Rwandan francs (RWF).
9. **Implementation Plan:** Outline the main stages of the project and the time needed for each stage.